

CPTG Native-Clock Structural Stress Test

MQN01 J004131.9-493704 (“Red Potato”)

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Abstract

MQN01 J004131.9-493704, the “Red Potato,” is remarkable because it is already massive, compact, and quiescent at spectroscopic redshift $z = 3.2496$, while residing in a gas-rich cosmic-web environment. Published measurements give $\sigma_\star = 268 \pm 20 \text{ km s}^{-1}$, a stellar-mass estimate $M_\star = 1.1_{-0.4}^{+0.4} \times 10^{11} M_\odot$, and a rest-frame optical half-light-radius estimate near 0.9 kpc under the source paper’s flat- Λ CDM $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ distance convention. Because that conventional normalization lies near the midpoint between the Planck-like acoustic and SH0ES-like luminosity scales, the distance-dependent source coordinates are converted using the arithmetic midpoint of the CPTG acoustic and luminosity distance coordinates at the measured redshift. This neutral conversion coordinate is not a new CPTG branch. It gives a central $R_{e,\text{mid}} = 0.891 \text{ kpc}$ and a leading-order distance-rescaled stellar mass $M_{\star,\text{mid}} \simeq 1.08 \times 10^{11} M_\odot$. Redshift selects the common native π -branch phase, while the acoustic and luminosity projections assign their own clock coordinates to that same phase. The clock does not drive $a_\star$: in this one-radius pressure-supported reduction, $a_\sigma = \sigma_\star^2/R_e$ is used as the estimator of the object’s structural acceleration scale. At the observed phase this gives $a_\star = 2.6126 \times 10^{-9} \text{ m s}^{-2}$, and the central Sérsic reconstruction gives Structural Mode $N = 3.09498$, corresponding to Early Spiral in the current Curvature Structure Mode Index (CSMI). Projecting the same support coordinate to normalized present comparison coordinates and reapplying the same structural estimator yields $N_{0,\pi} = 3.27710$, $N_{0,\text{ac}} = 3.27114$, and $N_{0,\text{lum}} = 3.27087$. The luminosity comparison gives $\Delta N = +0.17589$ or 5.683% relative to the observed-epoch state. As a fixed-source comparison-coordinate diagnostic, the early-to-late contrast in logarithmic $a_\star$ relaxation is 5.71 when rates are expressed per gigayear and 1.33 when expressed per unit N_π . This is consistent with a denser native-clock cadence in the early universe, but is not an observed descendant history. Relative to the source paper’s $H_0 = 70$ scale, the symmetric CPTG conversion changes the central radius by only about one percent and leaves the principal structural classification unchanged.

1 Introduction: Why the Red Potato Matters

At first glance, the Red Potato is simply a compact red galaxy. What makes it unusual is *when* it is observed and how mature it already appears at that time. At $z = 3.2496$, the galaxy contains about $1.1 \times 10^{11} M_{\odot}$ in stars, is only about a kiloparsec across at its half-light radius, and is already forming stars at a very low rate. In conventional galaxy-evolution language, it has assembled a large stellar mass, compactified, and largely quenched while the Universe is still very young [1].

The environment makes the object even more informative. The Red Potato lies in the dense MQN01 cosmic-web structure and is surrounded by a large reservoir of cool circumgalactic gas, yet its compact stellar body contains little detected cold molecular gas and remains quiescent. A nearby luminous active galaxy and its likely jet provide a plausible source of the unusually strong circumgalactic turbulence [1]. For cosmologists, the interest is therefore not simply that an early galaxy is massive. It is the combination of rapid early assembly, early quiescence, abundant surrounding fuel, and a feedback environment capable of preventing that fuel from readily returning to the galaxy. This combination makes the Red Potato part of the broader high-redshift massive-quiescent population that places pressure on the timing and efficiency of current galaxy-formation models [2, 3, 4].

The same properties make the galaxy unusually useful for CPTG. Its redshift supplies a well-defined cosmological phase, while its measured stellar dispersion, size, mass, and Sérsic structure provide the ingredients needed to measure the galaxy itself at that selected phase. This paper therefore asks a simple sequence of questions: where does the native clock place the Red Potato; what structural acceleration $a_{\star}$ is measured there; how is the solved CPTG field organized through Structural Mode N and CSMI; and how do those quantities compare when the same observation is expressed through present branch coordinates? A final diagnostic asks whether the galaxy’s unusually rapid early maturation is better organized by elapsed gigayears or by the larger amount of native geometric phase accumulated per gigayear in the early universe.

The next section defines that sequence carefully, separating the role of the cosmological clock from the role of the galaxy structural measurement.

2 Purpose, Test Definition, and Observed Inputs

The Red Potato provides a direct test of how CPTG joins cosmological phase to galaxy structure. It is treated as an ordinary bound galaxy observed through a redshift-dependent comparison layer, not as a system requiring a special high-redshift gravity law. The calculation separates cosmological placement from structural measurement:

$$\begin{array}{c}
 \underbrace{z \longrightarrow N_{\pi} \longrightarrow \text{selected epoch and branch coordinates}}_{\text{cosmological placement}} \\
 \Downarrow \\
 \underbrace{\text{galaxy structure} \longrightarrow a_{\star} \longrightarrow N \longrightarrow \text{CSMI}}_{\text{structural measurement and classification}}
 \end{array} \tag{1}$$

The cosmological geometry fixes phase first; the galaxy is then solved from its own source and support state [7, 8]. In particular, $a_\star$ has no independent clock. Redshift selects

$$a = \frac{1}{1+z}, \quad N_\pi = \ln a, \quad H_b dt_b = dN_\pi, \quad (2)$$

so equal N_π means equal geometric phase, not equal accumulated branch time.

At the selected Red Potato phase, the observed pressure-supported state supplies the one-radius support coordinate a_σ . The core framework does not define a_σ and $a_\star$ as universally identical; rather, this declared effective-radius reduction uses a_σ as the estimator of $a_\star$. Branch comparisons do not evolve $a_\star$ with the clock. Instead, the measured support coordinate is represented at a selected comparison phase a_x by

$$a_{\sigma,b}^{(x)} = a_\sigma^{(\text{RP})} \frac{E_b(a_x)}{E_b(a_{\text{RP}})}, \quad a_{\star,b}^{(x)} = a_{\sigma,b}^{(x)} \quad \text{within this one-radius reduction.} \quad (3)$$

For the normalized present comparison, $a_x = 1$ and $E_b(1) = 1$, so Eq. (3) reduces to $a_{\sigma,b}^{(0)} = a_\sigma^{(\text{RP})}/E_b(a_{\text{RP}})$. For the illustrative formation slice below, $a_x = a_{\text{form}}$. This is a fixed-source comparison map on the measured support coordinate, not an equation of motion $a_\star(t)$ and not a descendant history.

2.1 Observed inputs

Wang et al. provide the quantities needed for both parts of the test: redshift locates the common cosmological phase; stellar dispersion and effective radius supply the pressure-support measurement; and stellar mass plus Sérsic structure define the radial baryonic source [1]. Their quoted physical sizes and stellar mass adopt a flat Λ CDM distance convention with $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3$, and $\Omega_\Lambda = 0.7$. The value $70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ lies close to the midpoint between the CPTG acoustic and luminosity present endpoints,

$$\frac{H_{0,\text{ac}} + H_{0,\text{lum}}}{2} = 70.3238983676 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (4)$$

For a neutral source-unit conversion, this paper therefore uses the arithmetic midpoint of the CPTG acoustic and luminosity *distance coordinates* at the measured redshift. This is only a symmetric conversion coordinate between two established projections; it is not a third projected branch and does not alter the native cosmological phase.

For $b \in \{\text{ac}, \text{lum}\}$, define

$$D_{M,b}(z) = \frac{c}{H_{0,b}} \int_0^z \frac{dz'}{E_b(z')}, \quad D_{A,b}(z) = \frac{D_{M,b}(z)}{1+z}, \quad (5)$$

with $D_{L,b} = (1+z)D_{M,b}$. The symmetric conversion coordinate is

$$D_{A,\text{mid}}(z) \equiv \frac{D_{A,\text{ac}}(z) + D_{A,\text{lum}}(z)}{2}, \quad (6)$$

and analogously for $D_{L,\text{mid}}$. At $z = 3.2496$,

$$D_A^{\Lambda 70} = 1549.557 \text{ Mpc}, \quad D_{A,\text{ac}} = 1593.610 \text{ Mpc}, \quad D_{A,\text{lum}} = 1474.340 \text{ Mpc}, \quad (7)$$

so

$$D_{A,\text{mid}} = 1533.975 \text{ Mpc}, \quad s_D \equiv \frac{D_{A,\text{mid}}}{D_A^{A70}} = 0.9899441. \quad (8)$$

The published central radius estimate therefore maps as $0.9 \text{ kpc} \rightarrow 0.891 \text{ kpc}$. The direct midpoint of the published $0.13''$ and $0.11''$ angular radii gives 0.892 kpc under the same midpoint distance coordinate, so the two routes agree at the sub-percent level. For the stellar-mass normalization, a leading-order distance-only rescaling at fixed observed flux gives $M_\star \propto D_L^2$ and therefore $1.1 \times 10^{11} M_\odot \rightarrow 1.078 \times 10^{11} M_\odot$. This correction is only about two percent in mass and is far smaller than the quoted $\pm 0.4 \times 10^{11} M_\odot$ observational uncertainty; it is retained only to keep the source normalization in the same comparison coordinate. No claim is made that this approximate rescaling replaces a full stellar-population refit.

The central reconstruction therefore uses $M_{\star,\text{mid}} = 1.078 \times 10^{11} M_\odot$, $R_{e,\text{mid}} = 0.891 \text{ kpc}$, and $n = 2$. The directly measured F150W2 and F322W2 angular profiles are converted with $D_{A,\text{mid}}$ and retained as sensitivity cases.

Quantity	Adopted value	Role in test
Redshift	3.2496 ± 0.0006	Common native phase
Stellar dispersion	$268 \pm 20 \text{ km s}^{-1}$	Pressure-support measurement
Stellar mass	$(1.078 \pm 0.392) \times 10^{11} M_\odot$	CPTG midpoint-converted source normalization
Rest-frame optical radius	0.891 kpc	CPTG midpoint-converted central scale
F150W2 profile	$0.13''$, $R_{e,\text{mid}} = 0.967 \text{ kpc}$, $n = 1.5$	Blue-profile sensitivity
F322W2 profile	$0.11''$, $R_{e,\text{mid}} = 0.818 \text{ kpc}$, $n = 2.6$	Red-profile sensitivity
CGM extent	about 80 kpc	Environmental context

Table 1: Primary inputs used for the native-clock structural reconstruction after converting distance-dependent source estimates from the source paper’s $H_0 = 70$ convention to the symmetric midpoint of the CPTG acoustic and luminosity distance coordinates.

These inputs now permit the calculation to proceed in the required order: cosmological placement first, galaxy structural measurement second.

3 Native-Clock and Branch-Projection Reconstruction

This section converts the observed redshift into a CPTG location before any structural classification is attempted. The native π branch supplies the underlying geometric history. Acoustic and luminosity quantities are projections of that same history: they provide different observational clock coordinates at a common phase rather than separate universes or separate versions of the galaxy. Once that phase is fixed, the galaxy’s own kinematic and photometric measurements are used to determine $a_\star$.

3.1 Turned-back phase and branch clocks

The first task is geometric: *where on the CPTG history is the galaxy observed?* The native π branch and its two comparison projections are [7, 8]

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}, \quad \mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad \mathcal{T}_\pi = \frac{1}{\pi}, \quad (9)$$

$$E_{\text{ac}}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\text{ac}}}, \quad p_{\text{ac}} = 3 - \frac{\pi}{100}, \quad (10)$$

$$E_{\text{lum}}(a)^2 = \Omega_{\Lambda, \text{ac}} + \Omega_{m, \text{ac}} a^{-p_{\text{ac}}}, \quad \Omega_{m, \text{ac}} = \frac{p_{\text{ac}}}{3\pi}, \quad \Omega_{\Lambda, \text{ac}} = 1 - \Omega_{m, \text{ac}}. \quad (11)$$

The corresponding present branch and projection normalizations are

$$H_0^{(\pi)} = 69.4162507897, \quad H_{0, \text{ac}}^{\text{CPTG}} = 67.4777967351, \quad \text{km s}^{-1} \text{Mpc}^{-1}. \quad (12)$$

$$H_{0, \text{lum}}^{\text{CPTG}} = 73.17$$

The luminosity history is the data-conditioned comparison realization defined in the Pi-branch reference, not a second native cosmology [8].

For $z = 3.2496$,

$$a_{\text{RP}} = 0.235316, \quad N_{\pi, \text{RP}} = -1.446825, \quad (13)$$

with $E_\pi = 5.53750$, $E_{\text{ac}} = 4.90148$, and $E_{\text{lum}} = 4.87683$ at that same phase. Integrating $t_b(a) = \int_0^a da' / (a' H_b)$ gives the branch-clock coordinates in Table 2.

Clock coordinate	t_{RP} (Gyr)	t_0 (Gyr)
Native π branch	1.631554	12.754121
Exact acoustic projection	2.010934	13.885174
Luminosity comparison	1.864144	12.843057

Table 2: Branch-specific times at one common Red Potato phase.

3.2 Historical structural measurement

With the observation located, the calculation turns to the galaxy itself. For a pressure-supported system the observed effective-radius support coordinate is [7]

$$a_\sigma(R_e) = \frac{\sigma_\star^2}{R_e}. \quad (14)$$

The core framework treats a_σ as an observable support coordinate rather than as a universal definition of $a_\star$. In this declared one-radius reduction, $a_\sigma(R_e)$ is adopted as the estimator of the Red Potato structural acceleration scale. For the midpoint-converted central radius,

$$a_\star^{(\text{RP})} = a_\sigma^{(\text{RP})} = \frac{(268 \text{ km s}^{-1})^2}{0.890950 \text{ kpc}} = 2.6126 \times 10^{-9} \text{ m s}^{-2}. \quad (15)$$

The corresponding local curvature-transport radius is

$$R_c^{(\text{RP})} = \frac{48\pi c^2}{a_\star^{(\text{RP})}} = 168.12 \text{ Gpc}. \quad (16)$$

At fixed $R_{e,\text{mid}} = 0.890950 \text{ kpc}$, the quoted dispersion interval gives $a_\star^{(\text{RP})} = (2.237\text{--}3.017) \times 10^{-9} \text{ m s}^{-2}$.

3.3 Present comparison after branch projection

The key ordering is projection first, structural measurement second. Equation (3) gives the declared present comparison and reapplies the same one-radius $a_\sigma \rightarrow a_\star$ estimator. For the central Red Potato support value this gives

Present comparison coordinate	$E_b(a_{\text{RP}})$	$a_{\star,b}^{(0)} \text{ (m s}^{-2}\text{)}$	$R_{c,b}^{(0)} \text{ (Gpc)}$
Native coordinate	5.53750	4.7179×10^{-10}	930.96
Acoustic projection	4.90148	5.3301×10^{-10}	824.03
Luminosity comparison	4.87683	5.3571×10^{-10}	819.89

Table 3: Present comparison structural scales obtained by projecting the measured pressure-support coordinate first and then applying the same one-radius $a_\sigma \rightarrow a_\star$ estimator. The luminosity row is the primary observational comparison for the optical/near-infrared structural measurements used in this paper.

These are coordinate projections of the same measured support state, not an $a_\star(t)$ evolution law. The next section asks how the corresponding solved response is organized radially through Structural Mode N .

4 Structural Mode N and CSMI by Epoch

Structural Mode N is CPTG’s dimensionless summary of *where the solved gravitational response is organized across a galaxy*. It is neither a gravity-amplitude measure nor a visual morphology label. The solved radial field is converted into a structural weight; its weighted centroid defines an organization length λ ; and $N = R/\lambda$ measures that organization relative to the adopted outer readout radius. Larger N places the structural weight farther inward as a fraction of R . CSMI is the downstream catalog mapping of the solved continuous N , with no Hubble type used as an input. The dedicated Structural Mode study gives the full construction, sampled estimator, CSMI intervals, and primary-SPARC validation [6].

Structural Mode is distinct from the cosmological phase $N_\pi = \ln a$: N_π states *when* the galaxy is represented, whereas N states *how* its solved field is radially organized. For a

solved field $g(r)$,

$$C_g(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr} \right)^2, \quad \kappa_{\text{struct}} = 1 \text{ kpc}, \quad (17)$$

$$w(r) = \left| \frac{dC_g}{dr} \right|, \quad \lambda = \frac{\int_{R/5}^R r w(r) dr}{\int_{R/5}^R w(r) dr}, \quad N = \frac{R}{\lambda}. \quad (18)$$

For the central source, the reconstruction assumes spherical symmetry and a spatially constant stellar mass-to-light ratio, with the projected stellar-mass profile following the adopted $n = 2$ Sérsic light profile. The source normalization and radius are the midpoint-converted values above. A spherical Abel deprojection normalized to $M_\star$ gives

$$g_{\text{bar,RP}}(r) = \frac{GM_\star(< r)}{r^2}. \quad (19)$$

Because an ideal Sérsic profile has no finite edge, the declared readout is $R = R_{95} = 3.972 \text{ kpc}$. The same source and radial window are retained for every branch comparison.

For each structural scale a_s , the radial CPTG field is solved from

$$g(r) = g_{\text{bar,RP}}(r) + \frac{\sqrt{a_s g_{\text{bar,RP}}(r) + g_{\text{geom}}(r)}}{\left[1 + (g(r)/a_s)^{123/50} \right]^{11/50}}, \quad (20)$$

$$g_{\text{geom}}(r) = \frac{1}{2} a_s \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \quad R_c = \frac{48\pi c^2}{a_s}. \quad (21)$$

For every central and sensitivity reconstruction used here, the solved $C_g(r)$ is monotonically decreasing over the registered interval $R/5 \leq r \leq R$. Hence $w(r) = -dC_g/dr$ on that interval, and the continuous centroid can be evaluated with the exact endpoint identities

$$\begin{aligned} \int_{R/5}^R w(r) dr &= C_g(R/5) - C_g(R), \\ \int_{R/5}^R r w(r) dr &= \frac{R}{5} C_g(R/5) - R C_g(R) + \int_{R/5}^R C_g(r) dr. \end{aligned} \quad (22)$$

This endpoint-controlled evaluation is algebraically equivalent to the registered continuous definition for these monotone solutions, but avoids bias from a numerical grid that does not land exactly on $R/5$ and R .

At the observed Red Potato phase, $\lambda = 1.2835 \text{ kpc}$ and

$$N_{\text{RP}}^{(\text{hist})} = 3.09498, \quad \text{CSMI}_{\text{RP}}^{(\text{hist})} = \text{Early Spiral}, \quad (23)$$

using the current $3.05 < N \leq 3.30$ Early-Spiral interval. A coeval nearby observer would recover the same intrinsic $a_\star$, N , and CSMI; observational distance does not generate the present comparison values.

Applying the same radial solve after the branch projections gives

Structural state	a_* (m s ⁻²)	λ (kpc)	N	CSMI
Observed Red Potato	2.6126×10^{-9}	1.2835	3.09498	Early Spiral
Present native comparison	4.7179×10^{-10}	1.2122	3.27710	Early Spiral
Present acoustic comparison	5.3301×10^{-10}	1.2144	3.27114	Early Spiral
Present luminosity comparison	5.3571×10^{-10}	1.2145	3.27087	Early Spiral

Table 4: Observed-epoch and fixed-source present comparison-coordinate Structural Mode readouts. The present rows are not de-redshifted local views or descendant histories.

The primary luminosity comparison changes the central mode by $\Delta N_{\text{lum}} = 0.17589$, or 5.683%, while the native and acoustic comparison modes differ from it by less than 0.0063 and remain in the same CSMI interval.

The principal source uncertainties are summarized compactly in Table 5. The exact continuous N can also move near a CSMI boundary when the finite outer readout convention is changed; R_{95} is therefore a reconstruction convention, not a new CSMI rule.

Sensitivity case	Observed-epoch N /CSMI	Present $N_{0,\text{lum}}$ /CSMI
$\sigma_* = 248\text{--}288 \text{ km s}^{-1}$	3.0686–3.1218, Early Spiral	3.2628–3.2784, Early Spiral
$M_{*,\text{mid}} = 0.686 \times 10^{11} M_{\odot}$	3.00895, Intermediate Spiral	3.24133, Early Spiral
$M_{*,\text{mid}} = 1.470 \times 10^{11} M_{\odot}$	3.14660, Early Spiral	3.28479, Early Spiral
F150W2: $0.13''$, $R_{e,\text{mid}} = 0.967 \text{ kpc}$, $n = 1.5$	2.64495, Late Spiral	2.78452, Late Spiral
F322W2: $0.11''$, $R_{e,\text{mid}} = 0.818 \text{ kpc}$, $n = 2.6$	3.30930, Bulged Spiral	3.51099, Bulged Spiral

Table 5: Central observational and profile sensitivities after symmetric CPTG midpoint distance conversion. The mass rows are the source paper’s 0.7 and $1.5 \times 10^{11} M_{\odot}$ limits carried into the same symmetric CPTG midpoint distance convention; the profile rows use the directly measured angular radii. Each row is a fixed source description, not a morphology input or evolutionary track.

The CPTG reconstruction has now supplied a finite a_* , Structural Mode, and CSMI. The next section steps outside that calculation to explain why the same object is also a demanding galaxy-formation case for standard cosmology.

5 Challenge to Standard Λ CDM Galaxy Formation

The Red Potato combines several properties that are individually possible within standard Λ CDM galaxy formation but difficult to reproduce simultaneously at such an early epoch. The bounded claim is not that one galaxy falsifies cold dark matter; the stress lies in the timing, efficiency, and coexistence of the full observed state, within a broader high-redshift quiescent population that some simulations underproduce [2, 3, 4].

Rapid early assembly. By $z = 3.2496$, the Red Potato had already assembled a compact $\simeq 1.1 \times 10^{11} M_{\odot}$ stellar system. Rare massive halos are allowed, but the observation requires rapid halo growth together with unusually efficient early conversion of available baryons into stars. Other massive-galaxy samples can remain compatible with available Λ CDM halo abundances while still requiring very high baryon-to-star efficiencies [5].

Early quenching. The galaxy is already strongly quiescent, with a star-formation rate of only $4_{-2}^{+6} M_{\odot} \text{ yr}^{-1}$, no detected molecular-gas reservoir in the stellar body, and a molecular-gas fraction below about ten percent [1]. Quenching must therefore follow the rapid assembly very early. Standard mechanisms exist, but population-level success remains model dependent; IllustrisTNG- and ASTRID-based comparisons can underpredict the abundance of such quenched systems at $z \gtrsim 3$ by up to roughly two orders of magnitude [4].

Fuel without renewed star formation. The surrounding environment is not fuel poor: the galaxy lies in a dense cosmic-web node inside an approximately 80 kpc cool circumgalactic reservoir [1]. A successful model must therefore explain why accretion, cooling, or condensation remains inefficient despite abundant surrounding gas. This constrains coupled inflow, feedback, and circumgalactic physics rather than the dark-matter sector alone.

A possible preventive-feedback mechanism. AGN-ID2 lies about 48 kpc away in projection. X-ray and radio evidence is consistent with a jet directed into the Red Potato environment, making jet-driven turbulence a plausible mechanism for suppressing accretion and renewed star formation [1]. The association is suggestive rather than unique: the observations do not establish the jet as the sole cause of quiescence.

Taken together, these properties make the Red Potato a bounded but unusually informative stress case. Standard Λ CDM does not forbid such a galaxy; the challenge is to reproduce its rapid assembly, early quenching, continued quiescence in a fuel-rich environment, and relevant population abundance within one consistent formation history. CPTG does not replace the astrophysical microphysics of star formation, cooling, or feedback; its role here is the gravitational and structural accounting developed in the preceding sections.

6 Results, Interpretation, and Limitations

The primary numerical results are already collected in Tables 2, 3, and 4. In brief, the observed redshift places the Red Potato at $N_{\pi, \text{RP}} = -1.446825$ and $t_{\pi, \text{RP}} = 1.63155 \text{ Gyr}$ on the native history. Its midpoint-converted one-radius pressure-support estimator gives $a_{\star} = 2.6126 \times 10^{-9} \text{ m s}^{-2}$ and $N = 3.09498$ (Early Spiral). The primary luminosity present-coordinate comparison gives $a_{\star, \text{lum}}^{(0)} = 5.3571 \times 10^{-10} \text{ m s}^{-2}$ and $N_{0, \text{lum}} = 3.27087$, also Early Spiral. These are fixed-source coordinate comparisons, not evidence that redshift changes the intrinsic galaxy or that an unobserved descendant followed this path.

6.1 Native phase cadence and fixed-source relaxation diagnostic

The stellar-population reconstruction does not determine a unique birth instant. For a bounded diagnostic, this paper declares $t_{\text{lum, form}} \simeq 0.920 \text{ Gyr}$ as an illustrative early-formation slice on the same luminosity comparison clock; it is a chosen proxy, not an

observed birth time [1]. Inverting the clock gives

$$a_{\text{form}} = 0.145863, \quad z_{\text{form}} = 5.85575, \quad t_{\pi, \text{form}} = 0.77197 \text{ Gyr}. \quad (24)$$

Under the same fixed central source reconstruction, Eq. (3) with $b = \text{lum}$ and $a_x = a_{\text{form}}$ gives $a_{\star, \text{form}} = 5.2550 \times 10^{-9} \text{ m s}^{-2}$, and the endpoint-controlled structural solve gives $N_{\text{form}} = 2.96092$ (Intermediate Spiral). Formation to observation therefore spans 0.85958 Gyr, during which this diagnostic changes $a_{\star}$ by a factor of 2.0114 and raises N by 0.13406.

The relevant native cadence is

$$\frac{dN_{\pi}}{dt_{\pi}} = H_{\pi}(a), \quad \Delta N_{\pi} = \int H_{\pi} dt_{\pi} = \ln\left(\frac{a_2}{a_1}\right). \quad (25)$$

Here “tick” means a continuous increment dN_{π} , not a discrete event or a different physical kind of time.

State	N_{π}	t_{π} (Gyr)	H_{π} (Gyr^{-1})	Cadence / present
Formation proxy	−1.92509	0.77197	0.82605	11.64
Observed Red Potato	−1.44683	1.63155	0.39312	5.54
Present native coordinate	0	12.7541	0.07099	1

Table 6: Native phase cadence for the illustrative formation slice, observed Red Potato phase, and normalized present coordinate.

Formation to observation accumulates $\Delta N_{\pi} = 0.47826$, or 0.55639 Gyr^{-1} on average; observation to the normalized present native coordinate accumulates $\Delta N_{\pi} = 1.44683$, or 0.13008 Gyr^{-1} . The early interval therefore accumulates native phase 4.28 times faster per gigayear.

This normalization materially changes the apparent structural-relaxation contrast. Define the positive logarithmic relaxation magnitude by

$$\mathcal{R}_{a,t} = -\frac{\Delta \ln a_{\star}}{\Delta t_{\pi}}, \quad \mathcal{R}_{a,N_{\pi}} = -\frac{\Delta \ln a_{\star}}{\Delta N_{\pi}}, \quad (26)$$

where $\Delta \ln a_{\star} = \ln a_{\star,2} - \ln a_{\star,1}$ for a forward comparison from state 1 to state 2. The formation-to-observation values are 0.8130 Gyr^{-1} and 1.4612 , while the observation-to-present-coordinate fixed-source values are 0.14246 Gyr^{-1} and 1.0952 . Thus the early-to-late contrast is 5.71 when the logarithmic relaxation rates are expressed per gigayear and 1.33 when they are expressed per unit native phase. For Structural Mode, the corresponding contrast falls from about 9.86 to 2.31. Native phase therefore organizes the fixed-source comparison more uniformly than elapsed gigayears alone, but this diagnostic does not establish native cadence as the sole cause of galaxy evolution and does not reconstruct an observed descendant.

The principal limitations are observational rather than arithmetic: unresolved stellar kinematics, the aperture convention for $\sigma_{\star}$, joint uncertainty in $M_{\star}$, R_e , n , and axis ratio, and the finite outer readout required for an ideal Sérsic profile. The source paper reports distance-dependent physical quantities under a flat- Λ CDM $H_0 = 70$ convention; this paper

converts those estimates to the symmetric acoustic/luminosity midpoint distance before the structural solve, while recognizing that a full CPTG stellar-population refit could refine the approximate mass rescaling. Spatially resolved stellar kinematics would permit a fuller Jeans reconstruction and sharpen both a_* and N . The luminosity history also remains a data-conditioned comparison realization rather than an independently derived microscopic luminosity projector [8]. Actual descendant evolution—including mergers, accretion, stellar mass loss, size growth, and environmental change—is outside the scope of the fixed-source comparison.

The structural calculation is therefore complete within its stated assumptions. Section 14 closes the paper by returning to the observed astrophysical life of the galaxy itself.

14 The Life Branch of the “Red Potato”

The preceding sections reduce the Red Potato to the measurements needed for a structural calculation. This final section restores the object-level narrative: what the galaxy had already become, what its environment contains, and what may have helped keep it quiescent.

By $z = 3.2496$, the Red Potato is already massive, compact, and largely quiescent: $M_* \simeq 1.1 \times 10^{11} M_\odot$, $R_e \simeq 0.8\text{--}1.0$ kpc, and $\sigma_* = 268 \pm 20$ km s $^{-1}$. Strong Balmer and metal absorption together with a pronounced D4000 break indicate that a substantial older stellar population was already present. Wang et al.’s best-fit stellar-population history further suggests that star formation had begun declining a few hundred million years before observation, although the beginning of stellar assembly is not directly measured [1].

That maturation occurred in the unusually rich MQN01 node, where massive galaxies, active nuclei, and abundant gas coexist at $z \gtrsim 3$. Yet the Red Potato itself forms stars at only $4_{-2}^{+6} M_\odot \text{ yr}^{-1}$, more than an order of magnitude below the contemporaneous main sequence, and its stellar body is poor in cold molecular gas: CO(4–3) and Na D are not detected and the molecular-gas fraction is below about ten percent [1]. The galaxy is therefore quiescent without living in a fuel-poor environment.

The contrast is clearest in the circumgalactic medium. Cool $10^4\text{--}10^5$ K gas traced by Ly α spans roughly 80 kpc, while Ly α velocity dispersions reach about 400–500 km s $^{-1}$ at projected radii of 20–30 kpc. Within the galaxy, H α indicates a dispersion-dominated ionized component with a characteristic dispersion near 200 km s $^{-1}$ and no strong ordered velocity gradient [1]. Thus the system is gas rich on circumgalactic scales but comparatively gas poor and quiet in its compact stellar core.

AGN-ID2, only about 48 kpc away in projection, provides a plausible preventive-feedback mechanism. Its X-ray luminosity is approximately 10^{45} erg s $^{-1}$ in the 2–10 keV band; extended X-ray emission and steep-spectrum radio emission support the interpretation of a jet directed into the Red Potato environment. Wang et al. argue that this jet could maintain the elevated CGM turbulence and reduce the efficiency with which cool gas settles back onto the galaxy, while appropriately leaving room for additional gravitational interaction or earlier feedback episodes [1].

The resulting life picture is coherent: the Red Potato assembled a large stellar mass early,

entered a declining star-formation phase, and remained quiescent even though cool gas persisted around it. Its importance lies precisely in that separation between abundant external fuel and a mature, compact stellar body that no longer uses that fuel efficiently.

Data, Code, and Evidence Availability

All observational inputs used in this study are drawn from the cited literature, principally Wang et al. for MQN01 J004131.9-493704. The CPTG quantities reported here are deterministic outputs of the equations, source assumptions, and comparison coordinates stated in the manuscript. The numerical reconstruction includes the source-paper $H_0 = 70$ to symmetric CPTG acoustic/luminosity midpoint distance conversion, leading-order stellar-mass distance rescaling, Sérsic Abel deprojection, implicit radial field solution, endpoint-controlled continuous Structural Mode evaluation, branch-clock evaluation, and the formation/cadence diagnostics.

The calculation scripts, machine-readable numerical outputs, and compilation/audit records used to reproduce the reported values are maintained with the manuscript evidence package and are available from the author. Numerical outputs retain additional guard digits for reproducibility; the manuscript reports rounded values appropriate to interpretation and those guard digits should not be read as observational precision.

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